

Call & Response MIDI (v2.0.0)

Intelligent Algorithmic Phrase Generator & 303-Style Sequencer

By Viral Samples

Overview

Call & Response MIDI is a generative phrase synthesizer, performance sequencer, and MIDI composition utility designed to bring structured musical conversation to electronic music production.

Unlike conventional randomizers or arpeggiators that produce disconnected note sequences, Call & Response MIDI operates on the foundational principle of musical dialogue: an opening phrase (**Call**) establishes the motif, and a complementary counter-phrase (**Response**) delivers the harmonic, modal, or rhythmic resolution.

The result is focused, coherent musical hooks, acid basslines, leads, and ostinatos—generated in seconds and shaped with complete procedural control.

Key Features

1. Dual-Phrase Dialogue Architecture

- **Independent Call & Response Lanes:** Dedicated lanes with individual mute switches, octave transpositions, phrase inversion, and retrograde reversal.
- **Selective Mutation & Dice:** Dedicated per-lane DICE mutators with adjustable mutation depth percentage (**MUTATE %**), allowing subtle phrasing variations without destroying the original hook.
- **Phrase Locking:** Individual padlock controls freeze either lane to protect favorite melodies during global rolls.

2. Modal Engine & Procedural Composition

- **35 Musical Scales:** Full support for Diatonic, Pentatonic, Blues, Exotic, and Arpeggio pools with real-time automatic conforming on key or scale changes.
- **Four-Tab Inspector Panel:**
 - **Generate:** 40 musical styles, 24 motif lead-in hooks, 38 response strategies, mode routing, and cadence endings with individual vector locks.
 - **Shape:** Gate length scaling (0.25x to 2.0x), continuous velocity control (1 – 127), harmonic resolve, and overdrive.
 - **Articulation:** TB-303-style accent boost, portamento slide time, filter cutoff (60 Hz – 14 kHz), resonance, and envelope decay.
 - **Options:** Flexible channel routing (Combined, Split, or Layered) and DAW timeline synchronization modes.

3. Tactile Performance & Modern Interface

- **Fixed Aspect-Ratio Interface:** Scalable, high-contrast dark theme calibrated for studio monitors and laptops without distortion or clipped controls.
- **Direct-Access Top Ribbon:** Live value readouts for Octave, Key, Scale, Accidental (b/#), Steps, Rate, and Swing, controllable via dropdown menus or stepper arrows.
- **Tempo-Relative Playback Rate:** Switch seamlessly between Half speed (0.5×), Normal (1.0× 16th-note sync), and Double speed (2.0×).
- **Continuous Modulation Dragging:** Intuitive vertical click-and-drag editing for Swing (0% – 100%) and Velocity (1 – 127).
- **Nudge & Step Shifting:** Dedicated [<] **OFFSET** [>] buttons to advance or retard note patterns by single steps.

4. Interactive Piano Roll & Built-in 303 Preview Engine

- **Visual Color-Coded Matrix:** Lane-matched note blocks (Orange for Call, Teal for Response) with double-click note editing and step auditioning that automatically silences upon mouse release.
- **Repeatable Gate Resizing:** Click and drag the right edge of any note up to 4.0× step duration.
- **Integrated Analog-Modelled Synth:** Built-in 303-inspired monophonic synthesizer featuring a resonant 24 dB low-pass filter, snappy envelope decay, and saturation—enabling zero-latency previews without wiring auxiliary virtual instruments.

5. Seamless DAW Integration & Presets

- **Direct MIDI Drag & Drop:** Export generated phrases directly onto DAW MIDI tracks.
 - **8 Rapid-Access Preset Slots (A–H):** Single-click recall, keyboard shortcut saving, and non-destructive 24-step undo/redo history.
 - **Global Panic Button:** Instant All-Notes-Off reset across all 16 MIDI channels.
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Technical Specifications & System Requirements

Specification	macOS	Windows
Operating System	macOS 10.14 (Mojave) or later (Catalina, Big Sur, Monterey, Ventura, Sonoma, Sequoia)	Windows 10 or Windows 11 (64-bit)
Architectures	Universal Binary: Native Apple Silicon (<code>arm64</code> : M1, M2, M3, M4) & Intel (<code>x86_64</code>)	64-bit Intel / AMD x86-64
Plugin Formats	VST3, Audio Unit (AU v2), Standalone Application	VST3, Standalone Application (<code>.exe</code>)
Validation	100% Passed Apple <code>auval</code> (Audio Unit Validation Tool)	Standard VST3 Validator Compliant
DAW Compatibility	Ableton Live, Logic Pro, FL Studio, Studio One, Cubase, Reaper, Bitwig Studio	Ableton Live, FL Studio, Studio One, Cubase, Reaper, Bitwig Studio, Cakewalk
Installation	Dedicated <code>.pkg</code> installer with automated permissions & AU cache registration	Portable NSIS installer (<code>Setup.exe</code>) with static C++ runtime linking

Attributions

Call & Response MIDI stands on the shoulders of creative musical innovators:

- **NordI3**: Original concept and pioneering algorithmic dual-phrase musical dialogue.
- **SideBrain**: Renowned Max for Live implementation and workflow concepts.
- **Viral Samples**: Native cross-platform C++/JUCE architecture, modern UI, analog synth engine, and expanded feature set.